# Modelling and Stability Analysis in Hyperelastic Membranes

AE 494: B.TECH. PROJECT - II

Presented by
**Anuttar Jain – 22B0003**

Under the guidance of
**Professor Krishnendu Haldar**

# BTP-I Review

Modelling and Stability Analysis in Hyperelastic Membranes

# Past Work

- Analysed the formulation of Classical Plate theory and Foppl von-Karman Model. Observed their Advantages and disadvantages.

- Studied a more appropriate formulation for the modelling membranes under large deformation, given by Healey et al. (2013).

- Used COMSOL Multiphysics to numerically solve the system of PDEs and study the corresponding results.

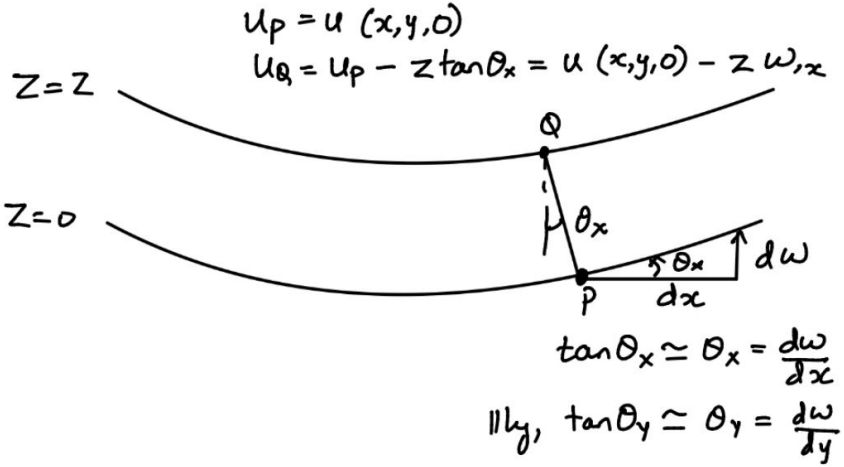
# Key Assumptions

## 1. Kirchhoff – Love Assumption

$$u_1(x, y, z) = u_1(x, y, 0) - z\frac{\partial w}{\partial x}\bigg|_{z=0}$$

$$u_2(x, y, z) = u_2(x, y, 0) - z\frac{\partial w}{\partial y}\bigg|_{z=0}$$

$$w(x, y, z) = w(x, y, 0)$$




## 2. Decomposition of Total Strain Energy

- Total Strain Energy Density can be decomposed into Membrane and Bending components under small thickness assumption

$$\Psi(\mathbf{E}, \mathbf{K}) = \Psi_m(\mathbf{E}) + \Psi_b(\mathbf{K})$$

# Large Deformation Model

$$\nabla \cdot [(\mathbf{I} + \nabla \mathbf{u})\mathbf{N}] + \mathbf{f}_m = \mathbf{0} \tag{1}$$

$$\nabla \cdot [\nabla \cdot \mathbf{M} + \mathbf{N}\nabla w] + f_b = 0 \tag{2}$$

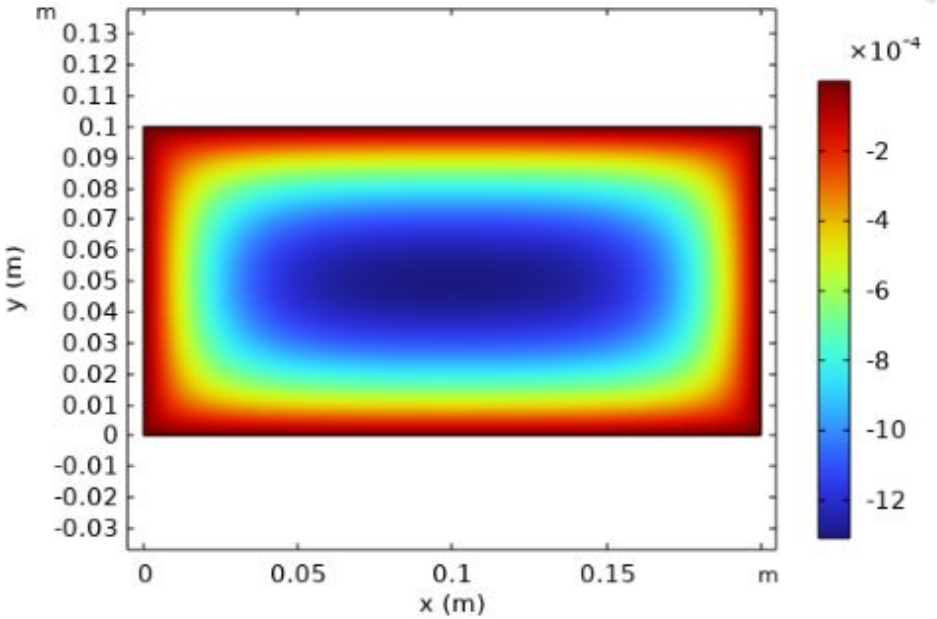
$$\mathbf{N} = \frac{\partial \Psi_m}{\partial \mathbf{E}} \tag{3}$$

$$\mathbf{M} = \frac{\partial \Psi_b}{\partial \mathbf{K}} \tag{4}$$

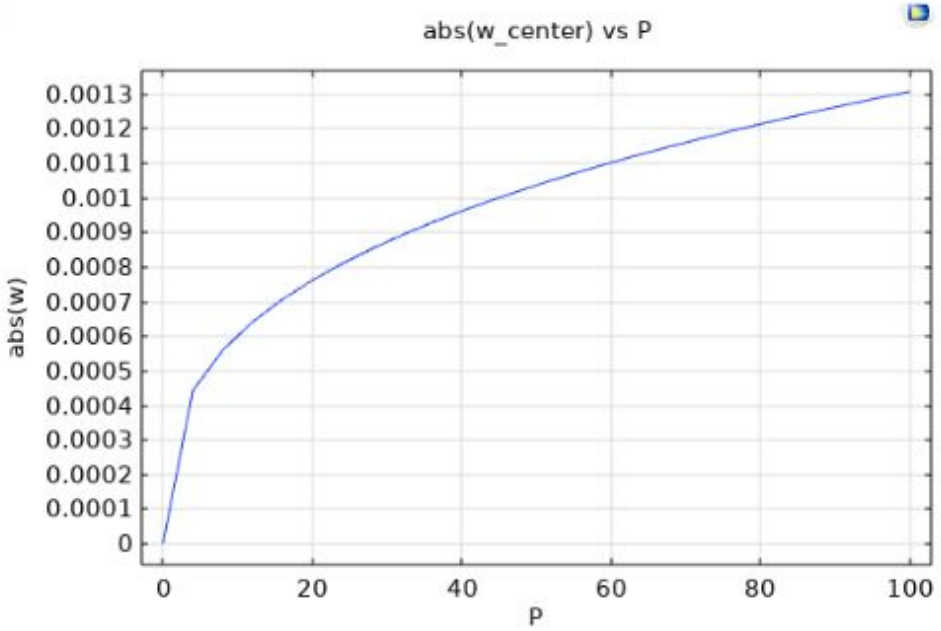
$$\mathbf{E} = \frac{1}{2}\left[\, \nabla \mathbf{u} + (\nabla \mathbf{u})^T + (\nabla \mathbf{u})^T(\nabla \mathbf{u}) + \nabla w \otimes \nabla w\right] \tag{5}$$

$$\mathbf{K} = -\nabla^2 w \tag{6}$$

# Plots of w(x,y) and |w| vs P




Plot of w(x,y)




$|w_{center}|$ vs P

# Spatial Stability Analysis

Modelling and Stability Analysis in Hyperelastic Membranes

# Perturbation Formulation

- Let u0, N0 be the equilibrium solution and their variations denote small perturbation about the equilibrium. Then we formulate the perturbation equation as follows -

$$\nabla \cdot [(\boldsymbol{I} + \nabla \tilde{\boldsymbol{u}})\tilde{\boldsymbol{N}}] = \boldsymbol{0}$$

$$\implies \nabla \cdot [(\boldsymbol{I} + \nabla \boldsymbol{u}_0 + \nabla \delta \boldsymbol{u})(\boldsymbol{N}_0 + \delta \boldsymbol{N})] = \boldsymbol{0}$$

$$\implies \nabla \cdot [(\boldsymbol{I} + \nabla \boldsymbol{u}_0)\boldsymbol{N}_0] + \nabla \cdot [(\boldsymbol{I} + \nabla \boldsymbol{u}_0)\delta \boldsymbol{N}] + \nabla \cdot [(\nabla \delta \boldsymbol{u})\boldsymbol{N}_0] + \nabla \cdot [(\nabla \delta \boldsymbol{u})\delta \boldsymbol{N}] = \boldsymbol{0}$$

$$\nabla \cdot [(\boldsymbol{I} + \nabla \boldsymbol{u}_0)\delta \boldsymbol{N}] + \nabla \cdot [(\nabla \delta \boldsymbol{u})\boldsymbol{N}_0] = 0$$

$$(\nabla^2 \delta \boldsymbol{u}) : \boldsymbol{N}_0 + (\nabla \delta \boldsymbol{u})(\nabla \cdot \boldsymbol{N}_0) + \nabla^2 \boldsymbol{u}_0 : \delta \boldsymbol{N} + (\boldsymbol{I} + \nabla \boldsymbol{u}_0)(\nabla \cdot \delta \boldsymbol{N})$$

$$\begin{bmatrix} N_{11} & 0 & 0 & 0 & 0 \\ 0 & N_{11} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta N_{11} \\ \delta N_{12} \\ \delta N_{22} \end{bmatrix}_{xx} + \begin{bmatrix} 2N_{12} & 0 & 0 & 0 & 0 \\ 0 & 2N_{12} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta N_{11} \\ \delta N_{12} \\ \delta N_{22} \end{bmatrix}_{xy} + \begin{bmatrix} N_{22} & 0 & 0 & 0 & 0 \\ 0 & N_{22} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta N_{11} \\ \delta N_{12} \\ \delta N_{22} \end{bmatrix}_{yy}$$

$$+ \begin{bmatrix} N_{11,x} + N_{12,y} & 0 & 1 + u_{1,x} & u_{1,y} & 0 \\ 0 & N_{11,x} + N_{12,y} & u_{2,x} & 1 + u_{2,y} & 0 \end{bmatrix} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta N_{11} \\ \delta N_{12} \\ \delta N_{22} \end{bmatrix}_{x}$$

$$+ \begin{bmatrix} N_{12,x} + N_{22,y} & 0 & 0 & 1 + u_{1,x} & u_{1,y} \\ 0 & N_{12,x} + N_{22,y} & 0 & u_{2,x} & 1 + u_{2,y} \end{bmatrix} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta N_{11} \\ \delta N_{12} \\ \delta N_{22} \end{bmatrix}_{y}$$

$$+ \begin{bmatrix} 0 & 0 & u_{1,xx} & 2u_{1,xy} & u_{1,yy} \\ 0 & 0 & u_{2,xx} & 2u_{2,xy} & u_{2,yy} \end{bmatrix} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta N_{11} \\ \delta N_{12} \\ \delta N_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{5.1}$$

# Perturbed System

$$\boldsymbol{Y} = (u_1, u_2, w, N_{11}, N_{12}, N_{22}, M_{11}, M_{12}, M_{22}, E_{11}, E_{12}, E_{22}, K_{11}, K_{12}, K_{22})^T$$

$$\boxed{\boldsymbol{A}\frac{\partial^2 \delta \boldsymbol{Y}}{\partial x^2} + \boldsymbol{B}\frac{\partial^2 \delta \boldsymbol{Y}}{\partial x \partial y} + \boldsymbol{C}\frac{\partial^2 \delta \boldsymbol{Y}}{\partial y^2} + \boldsymbol{D}\frac{\partial \delta \boldsymbol{Y}}{\partial x} + \boldsymbol{E}\frac{\partial \delta \boldsymbol{Y}}{\partial y} + \boldsymbol{F}\delta \boldsymbol{Y} = \boldsymbol{0}}$$

- System of 15 Second Order PDEs
- Quasilinear
- Behaviour is Hyperbolic or Elliptic depending on the load

# Cauchy Problem

$$\boldsymbol{A}\,\delta\boldsymbol{Y}_{,xx} + \boldsymbol{B}\,\delta\boldsymbol{Y}_{,xy} + \boldsymbol{C}\,\delta\boldsymbol{Y}_{,yy} = \boldsymbol{F}(x, y, \delta\boldsymbol{Y}, \delta\boldsymbol{Y}_{,x}, \delta\boldsymbol{Y}_{,y})$$

Let Γ be the cauchy curve over which the following data is specified

$$x = x_0(s), \quad y = y_0(s), \quad \frac{\partial \delta\boldsymbol{Y}}{\partial x} = \boldsymbol{p}_0(s), \quad \frac{\partial \delta\boldsymbol{Y}}{\partial y} = \boldsymbol{q}_0(s), \quad s_1 \le s \le s_2$$

$$\boldsymbol{p}_0' = \frac{\partial^2 \delta\boldsymbol{Y}}{\partial x^2} x_0' + \frac{\partial^2 \delta\boldsymbol{Y}}{\partial x \partial y} y_0'$$

$$\boldsymbol{q}_0' = \frac{\partial^2 \delta\boldsymbol{Y}}{\partial x \partial y} x_0' + \frac{\partial^2 \delta\boldsymbol{Y}}{\partial y^2} y_0'$$

# Characteristic Equation

For a unique Cauchy curve, there should be a unique solution to the 3 equations

$$\begin{vmatrix} \boldsymbol{A} & \boldsymbol{B} & \boldsymbol{C} \\ x_0'\boldsymbol{I} & y_0'\boldsymbol{I} & 0 \\ 0 & x_0'\boldsymbol{I} & y_0'\boldsymbol{I} \end{vmatrix} \neq 0$$

The following equation needs to be satisfied for existence of characteristics

$$\begin{vmatrix} \boldsymbol{A} & \boldsymbol{B} & \boldsymbol{C} \\ \dot{x}\boldsymbol{I} & \dot{y}\boldsymbol{I} & 0 \\ 0 & \dot{x}\boldsymbol{I} & \dot{y}\boldsymbol{I} \end{vmatrix} = 0$$

# Quadratic Eigenvalue Problem

Using Schur's Complement, and putting, $\lambda = -\dot{y}/\dot{x}.$, we get

$$\det\left(\lambda^2 \boldsymbol{A} + \lambda \boldsymbol{B} + \boldsymbol{C}\right) = 0.$$

$$M = \lambda^2 A + \lambda B + C, \quad \text{with eigenvectors } x.$$

$$L_1 = \begin{pmatrix} 0 & I \\ -C & -B \end{pmatrix} - \lambda \begin{pmatrix} I & 0 \\ 0 & A \end{pmatrix}, \quad \text{with eigenvector } z = \begin{pmatrix} x \\ \lambda x \end{pmatrix}$$

$$L_1 z = \begin{pmatrix} \lambda x - \lambda x \\ -Cx - \lambda Bx - \lambda^2 Ax \end{pmatrix} = \begin{pmatrix} 0 \\ -(\lambda^2 A + \lambda B + C)x \end{pmatrix} = 0.$$

# Dependence on Lambda

The behaviour of the system depends on $\lambda$:

(1) If $\lambda s$ are real and distinct $\Rightarrow$ Distinct Characteristics exist $\Rightarrow$ **Hyperbolic** Equation.

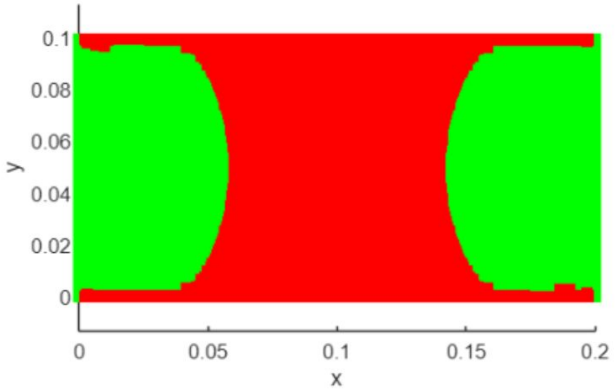
(2) If $\lambda s$ are real and repeated $\Rightarrow$ a Single Characteristic exists $\Rightarrow$ **Parabolic** Equation.

(3) If $\lambda s$ are non-real $\Rightarrow$ No Real Characteristics exist $\Rightarrow$ **Elliptic** Equation.

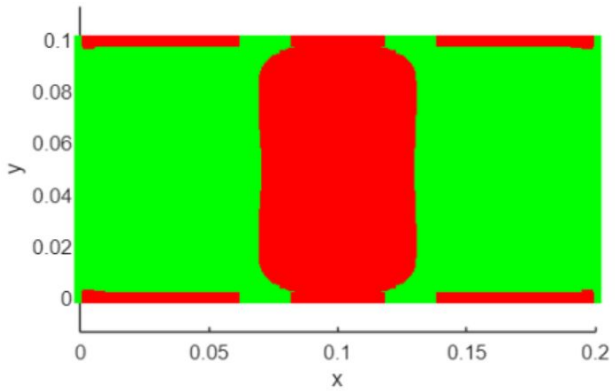
If even one of Re(λ) > 0 => Unstable Equilibrium

By varying loads, we can find the point of loss of stability in the system using λ. The load at which system turns from Elliptic to Hyperbolic marks this point.

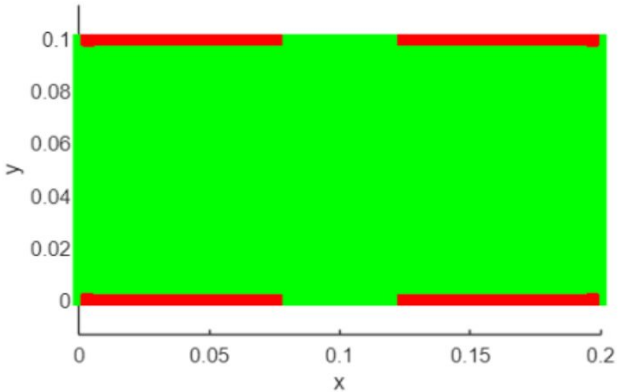
# Numerical Results




$\epsilon$ = 1e-4




$\epsilon$ = 0.1




$\epsilon$ = 0.2

Stability is regained at a high enough Applied Strain => Isola-centre Bifurcation!

# Path Following Methods

Modelling and Stability Analysis in Hyperelastic Membranes

# Terminologies

## Residual Function

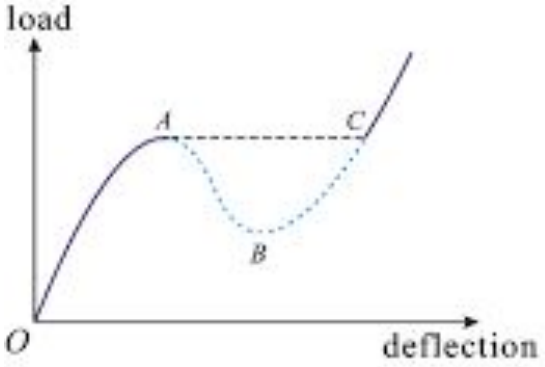
$$\boldsymbol{R} := \boldsymbol{R}(\boldsymbol{u}, \lambda) = \boldsymbol{f}_{internal} - \boldsymbol{f}_{external}$$

$$\boldsymbol{f}_{internal} = \boldsymbol{f}_{internal}(\boldsymbol{u})$$

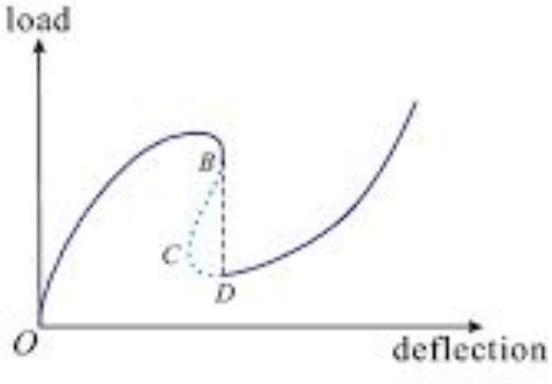
$$\boldsymbol{f}_{external} = \lambda \boldsymbol{f}_0$$

Ideally, $\boldsymbol{R}(\boldsymbol{u}, \lambda) = \boldsymbol{0}$ at equilibrium and represents a system of nonlinear equations.

## Load – Displacement Curve




(1) snap-through




(2) snap-back

# Newton–Rhapson Method

## Update Rule

$$\Delta \boldsymbol{u}^i = -\left( \left. \frac{\partial \boldsymbol{R}}{\partial \boldsymbol{u}} \right|_{\boldsymbol{u}^i} \right)^{-1} \boldsymbol{R}(\boldsymbol{u}^i, \lambda)$$

## Drawbacks

- Cannot proceed beyond limit points, i.e. points at which The tangential stiffness matrix is singular.
- Cannot Plot inside the Postbuckling Regime.

# Path Following Methods (Arclength Method)

## Update Rule

$$\begin{pmatrix} \Delta \boldsymbol{u}^i \\ \Delta \lambda^i \end{pmatrix} = \begin{pmatrix} \partial_{\boldsymbol{u}} \boldsymbol{R} & \partial_\lambda \boldsymbol{R} \\ \partial_{\boldsymbol{u}} g & \partial_\lambda g \end{pmatrix}^{-1} \begin{pmatrix} -\boldsymbol{R}(\boldsymbol{u}^i, \lambda^i) \\ -g(\boldsymbol{u}^i, \lambda^i) \end{pmatrix}$$

- g(u, λ) is a constraint relation.
- It is added to introduce another equation to update λ as well.
- We obtain different arclength methods for different g(u, λ).

# Crisfield Method

$$g(\boldsymbol{u}^i, \lambda^i) = \sqrt{(\boldsymbol{u}^i - \boldsymbol{u}^{i-1})^T(\boldsymbol{u}^i - \boldsymbol{u}^{i-1}) + (\lambda^i - \lambda^{i-1})^2}$$

## Advantages

- Depends upon only the previous value to find the next update.
- Uses constant arclength to cut the load displacement curve and returns the intersection point as the next iterate step.

## Drawbacks

- Too simplistic.
- Unable to find a feasible solution for complicated, large and sparse finite element models
- We need a more complicated method for our postbuckling analysis.

# Incremental Arclength with Cubic Extrapolation

Using the last 4 converged solution vectors **$X^0$, $X^1$, $X^2$, $X^3$** where: $X^i = (u^i, \lambda^i)$

### Extrapolation Formula

$$X^4 = \sum_{i=0}^{3} \left( \prod_{j=0, j\neq i}^{3} \frac{y_4 - y_j}{y_i - y_j} \right) X^i$$

### Iterative Definitions

$$y_0 = 0$$

$$y_1 = A_1$$

$$y_2 = y_1 + A_2$$

$$y_3 = y_2 + A_3$$

$$y_4 = y_3 + A_3$$

$$A_i = \left| X^i - X^{i-1} \right|$$

# Eigenmode Perturbation Method

If multiple bifurcating branches exist, the method stays on the initial branch and cannot switch between them. We need a suitable **branch switching method**.

## Bifurcation Analysis

At a bifurcation point, the tangential stiffness matrix becomes **singular** because multiple branches coexist.

Each branch is uniquely identified by a **basis eigenvector** of the null space.

## Perturbation & Switching

We perturb the solution at the bifurcation point along the direction of the eigenvector of our desired branch.

This forces the arclength method to **follow the specific desired branch**.

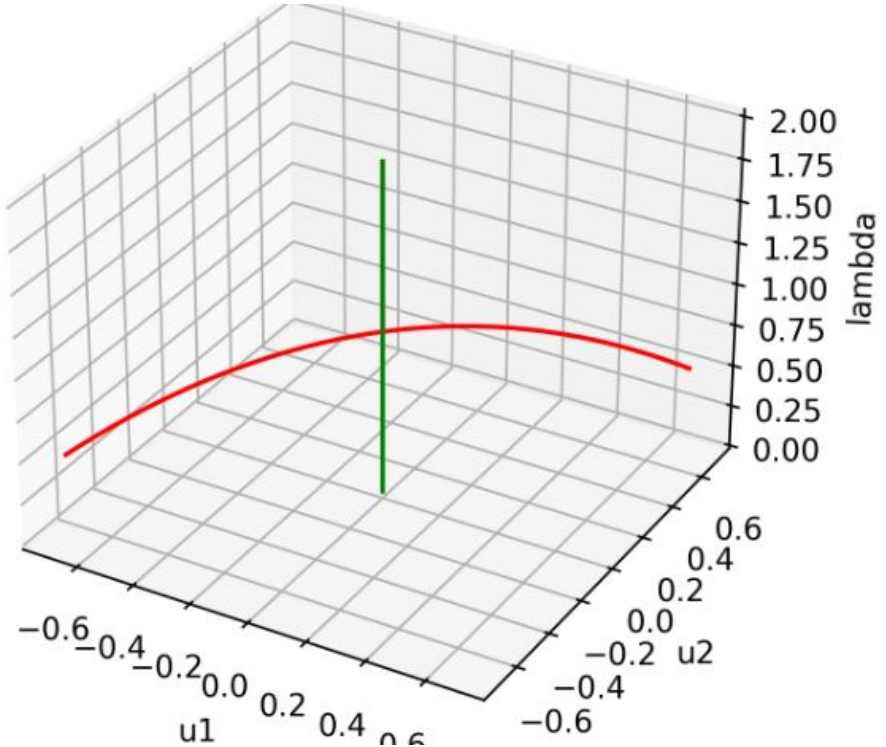
# Illustration Example

$$\boldsymbol{R}(\boldsymbol{u},\lambda) = \begin{pmatrix} (\lambda-1)u_1 + u_1^3 \\ u_2 - u_1 \end{pmatrix}$$

$$\boldsymbol{K_T}(0,0,\lambda) = \begin{pmatrix} \lambda - 1 & 0 \\ -1 & 1 \end{pmatrix}$$

**Key Characteristics:**

- KT is singular @ λ = 1.
- 2 Stable Branches
- 1 Unstable Branch (Not Shown)



# Magneto-hyperelasticity

Modelling and Stability Analysis in Hyperelastic Membranes

# Magneto-elastic Materials

- Smart materials in which mechanical deformation and magnetic fields are coupled.
- Applying a magnetic field can affect the stiffness of the material, thereby altering the buckling points, elastic regimes, yield point, etc.
- Typically made by embedding ferromagnetic particles (e.g., iron or magnetite) into a soft polymer matrix (like silicone/PDMS) and curing it, often under a magnetic field to control particle alignment.
- Our goal is to analyse the behaviour of thin membranes of magneto-elastic materials under large deformation conditions.

# Formulation Procedure

1. Define the mathematical framework (model description)
2. Obtain Kinematical Relations (Strain Tensors, etc.)
3. Decoupling the Total Strain Energy Density
4. Evaluate the expression for 2D Bending Strain Energy Density
5. Evaluate the expression for 3D Membrane Strain Energy Densities
6. Evaluating Stresses
7. Enforcing Plane Stress condition ($\mathbf{P}_{33} = 0$)
8. Reduce the 3D membrane stress tensor into 2D
9. Formulate Action Integral & find Equilibrium equations + Natural BCs

# Invariants and Compressibility Effects

1. $I_1 = trC$
2. $I_2 = \frac{1}{2}[(trC)^2 - tr(C^2)]$
3. $I_3 = det(C)$
4. $I_4 = H \cdot H$

$$\overline{F} = J^{-1/3}F$$

$$\overline{C} = J^{-2/3}C$$

$$\overline{I_1} = J^{-2/3}I_1$$

$$\overline{I}_2 = J^{-4/3}I_2$$

$$\overline{I}_3 = 1 \qquad (\because \text{Isochoric})$$

# Strain Energy Density per unit Area

$$\Psi^{3D}_{Total} = \Psi^{3D}_{Isochoric} + \Psi^{3D}_{Volumetric} + \Psi^{3D}_{Magneto-elastic} + \Psi^{3D}_{Magneto-elastic}$$

$$\Psi^{3D}_{Isochoric} = h[c_1(\bar{I}^{3D}_1 - 3) + c_2(\bar{I}^{3D}_2 - 3)]$$

$$\Psi^{3D}_{Volumetric} = h\left[\frac{1}{2}\kappa(J^{3D} - 1)^2\right]$$

$$\Psi^{3D}_{Magneto-elastic} = h[\hat{c}_2(\bar{I}^{3D}_2 - 3)]$$

$$\Psi^{2D}_{Bending} = \frac{Yh^3}{24(1-\nu^2)}\left[\nu(\mathrm{tr}\mathbf{K}^{2D})^2 + (1-\nu)\mathbf{K}^{2D} : \mathbf{K}^{2D}\right]$$

# Magneto-elastic Coupling Coefficients

$$\hat{c}_2(\zeta_2(I_4), \eta_2(I_4)) = \zeta_2 \tan^{-1}(\eta_2 \sqrt{I_4})$$

$$\zeta_2 = b_1 \ln(b_2 \sqrt{I_4} + 1)$$

$$\eta_2 = a_1 (e^{a_2 \sqrt{I_4}} - 1)$$

# Relation between 2D and 3D Quantities

$$\mathbf{F^{3D}} = \begin{pmatrix} \mathbf{F^{2D}} & \mathbf{0} \\ \mathbf{0} & \lambda_3 \end{pmatrix}$$

$$\mathbf{C^{3D}} = \begin{pmatrix} \mathbf{C^{2D}} & \mathbf{0} \\ \mathbf{0} & \lambda_3^2 \end{pmatrix}$$

$$J^{3D} = J^{2D} + \lambda_3$$

$$I_1^{3D} = I_1^{2D} + \lambda_3^2$$

$$I_2^{3D} = I_2^{2D} + I_1^{2D}\lambda_3^2$$

$$I_3^{3D} = I_3^{2D}\lambda_3^2$$

# Plane Stress Condition

$$\boldsymbol{S} = 2\frac{\partial \Psi_{membrane}}{\partial \boldsymbol{C}} = 2\frac{\partial \Psi_{isochoric}}{\partial \boldsymbol{C}} + 2\frac{\partial \Psi_{volumetric}}{\partial \boldsymbol{C}} + 2\frac{\partial \Psi_{magneto-elastic}}{\partial \boldsymbol{C}}$$

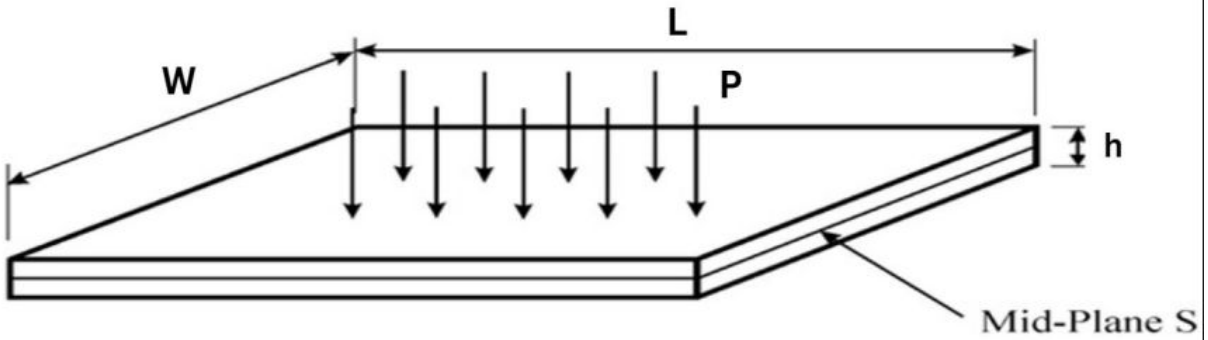
$$\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$$

$$\frac{kJ^2}{4}\lambda_3^{10/3} - \frac{kJ}{4}\lambda_3^{7/3} + \frac{1}{3}(c_2 + \hat{c}_2)J^{-4/3}I_1\lambda_3^2 - c_1 J^{-2/3}I_1\lambda_3^{2/3} - \frac{2}{3}(c_2 + \hat{c}_2)J^{-4/3} = 0$$

We obtained a $10^{th}$ order polynomial equation in $\mu = \lambda^{1/3}$, solving which, we can obtain a suitable value of $\lambda_3$ that will satisfy our plane stress condition.

# Problem Description

A thin cuboidal hyperelastic membrane made up of a magneto-active polymer matrix and having dimensions L x W x h, is kept such that all boundary edges are kept fixed. A uniform load P acts over the surface of the membrane in the downward direction.




**Body Forces**

$\mathbf{f}_m = \mathbf{0}, \qquad f_b = -P$

**Natural Boundary Conditions**

$\mathbf{n} \cdot \mathbf{Mn} = 0 \qquad$ along all edges

**Dirichlet Boundary Conditions**

$$\mathbf{u}(0, y) = \mathbf{u}(L, y) = \mathbf{u}(x, 0) = \mathbf{u}(x, W) = \mathbf{0} \qquad \forall\, x, y$$

$$w(0, y) = w(L, y) = w(x, 0) = w(x, W) = 0 \qquad \forall\, x, y$$

# Parameter Values

| Name | Value | Description |
|---|---|---|
| L | 0.2 m | Length |
| W | 0.1 m | Width |
| h | 0.002 m | Thickness |
| $\mu_0$ | $4\pi \times 10^{-7}$ H/m | Magnetic Permeability |
| $c_1$ | 108 kPa | Mooney-Rivlin Constant 1 |
| $c_2$ | 137 kPa | Mooney-Rivlin Constant 2 |
| $a_1$ | 4.97 m/A | Magneto-elastic coupling constant 1 |
| $a_2$ | $9.147\mu_0$ m/A | Magneto-elastic coupling constant 2 |
| $b_1$ | 174.685 kPa | Magneto-elastic coupling constant 3 |
| $b_2$ | $13.024\mu_0$ m/A | Magneto-elastic coupling constant 4 |
| $\nu$ | 0.49 | Poisson Ratio |
| $\mu$ | $2(c_1 + c_2)$ | Shear Modulus |
| $Y$ | $2\mu(1 + \nu)$ | Young's Modulus |
| $\kappa$ | $Y/(3(1 - 2\nu))$ | Bulk Modulus |

[2] Garai & Haldar

# $w_{Center}$ vs $H_x$ @ P = 1 kPa

